



The association of intestinal microbiota diversity and outcomes of allogeneic hematopoietic cell transplantation: a systematic review and meta-analysis

Shufen Wang^{1,2} · Xiaoyan Yue^{1,2} · Hongyu Zhou^{1,2} · Xu Chen^{1,2} · Huiqiao Chen^{1,2} · Liangning Hu¹ · Wenjue Pan¹ ·
Xiujie Zhao¹ · Haowen Xiao^{1,2} 

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Abstract

Growing evidence suggests that highly intestinal microbiota diversity modulates host inflammation and promotes immune tolerance. Several studies have reported that patients undergoing allo-HSCT have experienced microbiota disruption that is characterized by expansion of potentially pathogenic bacteria and loss of microbiota diversity. Thus, the primary aim of this meta-analysis was to determine the association of intestinal microbiota diversity and outcomes after allo-HSCT, and the secondary aim was to analyze the associations of some specific microbiota abundances with the outcomes of allo-HSCT. Electronic databases of Pubmed, Embase, Web of Science, and Cochrane Library were searched from inception to August 2023, and 17 studies were found eligible. The pooled estimate suggested that higher intestinal microbiota diversity was significantly associated with overall survival (OS) benefit (HR = 0.66, 95% CI: 0.55–0.78), as well as decreased risk of transplant-related mortality (HR = 0.56, 95% CI: 0.41–0.76), and lower incidence of grade II–IV aGVHD (HR = 0.41, 95% CI: 0.27–0.63). Furthermore, higher abundance of *Clostridiales* was associated with a superior OS (HR = 0.40, 95% CI: 0.18–0.87), while higher abundance of *Enterococcus* (HR = 2.03, 95% CI: 1.55–2.65), *γ-proteobacteria* (HR = 2.82, 95% CI: 1.53–5.20), and *Candida* (HR = 3.80, 95% CI: 1.32–10.94) was an adverse prognostic factor for OS. Overall, this meta-analysis highlights the protective role of higher intestinal microbiota diversity on outcomes after allo-HSCT during both pre-transplant and post-transplant periods. Some specific microbiota can be useful in the identification of patients at risk of mortality, offering new tools for individualized pre-emptive or therapeutic strategies to improve allo-HSCT outcomes.

Keywords Intestinal microbiota diversity · Overall survival · Transplant-related mortality · Allo-HSCT · GVHD

Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a potentially curative treatment for a variety of hematologic malignancies. The major complications after allo-HSCT, including relapse, graft-versus-host disease (GVHD), infection, and toxic effects to organs, limit the broader applicability of allo-HSCT [1–3]. Factors that can

decrease the risk of these adverse outcomes could contribute to important improvements in allo-HSCT efficacy.

The intestinal microbiota plays an important role in shaping host immunity immune reaction. The interaction between the human host and gut microbiota requires a delicate balance, which must be actively maintained to achieve a healthy and stable state. Once the gut microbiota is disrupted and imbalanced, it can lead to the loss of various functions of the host, such as barrier function loss, inflammation, and immune function loss, thereby inducing diseases [4]. In allo-HSCT patients, pre-treatment regimens before transplantation (including chemotherapy, radiation therapy, immunotherapy, and broad-spectrum antibiotics), as well as activation of allogeneic T cells after transplantation, can damage intestinal epithelial cells and alter intestinal microbial composition, leading to gut microbiota dysbiosis and decrease of bacterial diversity [5].

✉ Haowen Xiao
haowenxiaoxiao@zju.edu.cn

¹ Department of Hematology, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, No. 3 Qingchun East Rd, Hangzhou 310016, Zhejiang Province, People's Republic of China

² Institute of Hematology, Zhejiang University, Hangzhou, Zhejiang Province, People's Republic of China

In recent years, there have been growing evidences that the intestinal microbiota diversity is related to clinical outcomes of allo-HSCT [6–9]. The results, however, are not entirely consistent. Several studies have reported that higher microbiota diversity was associated with increased overall survival (OS) [6, 7], while others did not observe any significant association between the baseline intestinal microbiota diversity and OS [8, 10].

The healthy gut is inhabited by a diverse community of mostly anaerobic bacteria, and a hallmark of microbial dysbiosis involving the expansion of facultative anaerobic bacteria observed in many diseases [11]. Particularly, high relative abundances of *Enterobacteriaceae* and *Enterococcus* were associated with a decreased OS and an increased GVHD-related mortality [12–14]. In contrast, *Clostridiales* are associated with an improved outcome: reduced mortality for *Lachnospiraceae* [14], less GVHD for *Blautia* [15], and even decreased relapse rate for *Eubacterium limosum* [16].

Given lacking of systematic reviews and/or meta-analyses in this field, we performed this meta-analysis to determine the association of intestinal microbiota diversity and outcomes after allo-HSCT, including OS, transplant-related mortality (TRM), aGVHD, cGVHD, and relapse. More importantly, since the microbiota compositions of the majority of patients at HSCT have been changed and distinct from those of healthy volunteer because of conditioning regimens prior-HSCT [17], we evaluated the association of intestinal microbiota diversity before transplantation (pre-transplant) or after transplantation (post-transplant) separately. Furthermore, we analyzed the association of high abundance of *Clostridiales*, *Enterococcus*, *γ-proteobacteria*, and *Candida* with OS after allo-HSCT.

Methods

Registration and protocol

The systematic review protocol was registered on the “International Prospective Register of Systematic Review” under CRD42023403977. Moreover, the review was conducted and written as required by the “Preferred Reporting Items for Systematic Reviews and Meta-Analyses” (PRISMA) 2020 guideline [18].

Search strategy

The Pubmed, Embase, Cochrane library, and Web of Science database search was performed for articles involving human subjects that are in English from inception until August 11, 2023. We used a search strategy that included truncated free text and exploded MeSH terms relevant to “microbiota,” “mycobiome,” “bacteria,” “fungi,” “hematopoietic cell

transplantation,” “stem cell transplantation,” “bone marrow transplantation,” “diversity,” “abundance,” and “domination,” and more detailed search strategy are displayed in supplementary table S1. We identified additional articles by manually searching the reference lists from recent reviews, the extracted papers and database of Clinical Trials.

Study selection

All the records, including abstracts, were imported to End-Note X9 (Clarivate Analytics, Toronto, Canada). Duplicate records were first removed. The inclusion criteria are as follows: (1) assessed the association between microbiota and clinical outcomes in patients undergoing allo-HSCT; (2) had reported a hazard ratio (HR) or risk ratio (RR) and 95% confidential intervals (CI) for the associations or Kaplan–Meier plots from which we can estimate the HR and 95%CI. We excluded articles that fulfilled the following criteria: (1) non-English studies; (2) reviews, meta-analyses, book chapters, clinical guidelines, clinician perspectives, editorial, note, comment, case reports; (3) animal studies, and in vitro studies; (4) auto-HSCT studies; (5) non-gut microbiota; (6) trial registry record; (7) abstract, reviews, letters without original study, secondary studies, and studies that HR or 95% CI was unavailable in papers.

To identify eligible studies, we used a twostep selection process. Two independent investigators (ZHY, YXY) conducted an initial screening of all titles or abstracts and then evaluated all potentially relevant articles based on full text reviews.

Data extraction and quality assessment

Data extraction was conducted with a standardized data collection form two authors (WSF, CHQ) independently. We used hazard ratios or risk ratio and 95% confidence intervals (CI) as a measure of associations. The primary exposure variable was the diversity of intestinal microbiota, but we also assessed the association of some specific microbiota abundance with OS independently. Patients were stratified into higher-diversity and lower-diversity groups according to the median diversity value. Outcomes of interest in this study were OS, TRM, aGVHD, cGVHD, and relapse as described by Peled et al. [6].

We recorded the following characteristics in the identified studies: first author, publication year, study design, sample size of the cohort, country, donor types, conditioning regimens, age at entry, assessment method of intestinal microbiota diversity, categories of microbiota diversity, time of fecal sample collection, and variables that entered into the multivariable model as potential confounders. Regarding inclusion of studies and interpretation of data, a third investigator (XHW) was consulted to resolve discrepancies.

Any disagreements were settled through consensus with all three authors.

Statistical methods

We used Review Manager (RevMan) software version 5.4 (Cochrane, Copenhagen, Denmark) to analyze the data. For the effect of study-level variables, a p -value of < 0.05 will be regarded statistically significant. The multivariate-adjusted risk estimates were selected if they were reported in the original article; otherwise, the unadjusted risk estimates were calculated using the original data. When HR was unavailable directly from the articles, the Kaplan–Meier curve was evaluated by Engauge Digitizer version 12.1, and HRs and 95%CI were estimated using Richard Steven's excel workbook [19]. Using the Cochran's Q -test (p -value < 0.1 regarded significant) and I^2 ($> 50\%$ denoting substantial heterogeneity) statistics, we investigated the potential source of heterogeneity related to both methodological and clinical aspects of the studies. For evidence of $I^2 < 50\%$, we used a fixed-effects model for analysis; for studies with substantial heterogeneity, we used a random-effects model instead [20]. Because there were fewer than ten papers for each outcome, the publication bias was not addressed. Sensitivity analysis was performed to evaluate the effect of a particular study

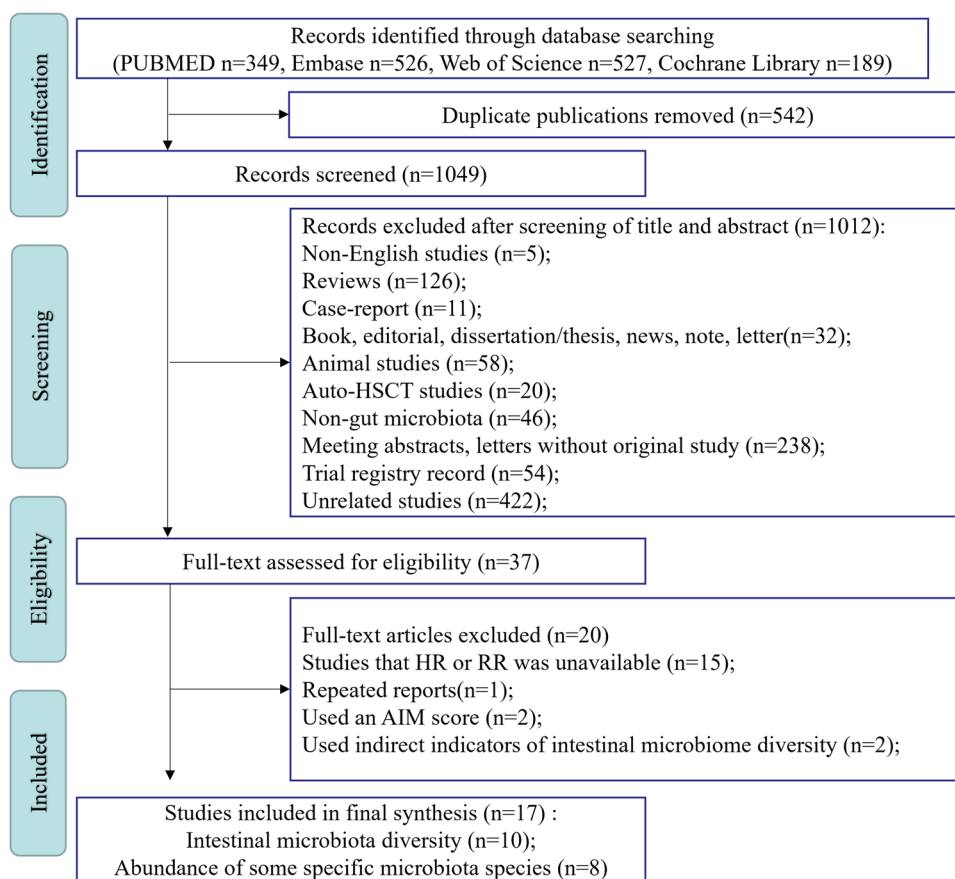
on the overall results by deleting one study at a time and combining the effect values of the remaining studies. We assessed study quality with the Newcastle–Ottawa quality assessment scale, providing a score range between 0 and 9 (≤ 6 , 7–8, and 9 indicate high, medium, and low risk of bias, respectively). We derived a score that summarized eight aspects of each study.

Results

Study search and study characteristics

Figure 1 shows study selection process and results from the literature search. We identified 349 articles from the Pubmed database, 526 articles from the Embase database, 527 articles from Web of Science, and 189 articles from Cochrane Library. After exclusion of duplicates and papers that did not meet the inclusion criteria, we obtained 37 full articles of potentially relevant studies. After full text reviews, we excluded twenty: one article [21] with data from the cohort included in another study [22]; two studies [23, 24] used an accumulated intestinal microbiota (AIM) score, which scored the inverse Simpson index and the abundance of *Lachnospiraceae*, *Peptostreptococcaceae*,

Fig. 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart of the study selection process



Erysipelotrichaceae, and *Enterobacteriaceae* as 1 or 0, to predefine microbiota; two studies used indirect indicators of intestinal microbiota diversity, such as urinary 3-indoxyl sulfate (3-IS) [25] and serum concentration of islet-derived 3 alpha (Reg3 alpha) [26]; 15 articles [10, 27–40] with insufficient data for association between intestinal microbiota diversity and relevant outcomes. Finally, 17 articles were selected for further data extraction and analysis. Ten articles [6–9, 41–46] examined the association between intestinal microbiota diversity and clinical outcomes after allo-HSCT, eight articles [8, 12, 14, 15, 47–50] reported the association of the abundance of some specific microbiota with clinical outcomes of allo-HSCT, and one of them [8] showed the association of both intestinal microbiota diversity and the abundance of some specific microbiota with the outcomes.

The characteristics of the included studies are shown in Table S2 and Table S3. In total, patients undergoing allo-HSCT were enrolled between 2009 and 2020. Ten studies collected fecal samples after transplantation during day 0 to day 180 (defined as post-transplant) [7, 12, 15, 41–43, 45, 46, 49, 50], four studies collected fecal samples before transplantation during day 30 to day 1 (defined as pre-transplant) [8, 9, 14, 44], and three studies examined both pre-transplant and post-transplant fecal samples [6, 47, 48]

Assessment of study quality yielded an average score of 7.4, and fifteen studies had a score of 6.5 or above (see Table S4).

The association of intestinal microbiota diversity and outcome indicators

Overall survival

Data to calculate the association of intestinal microbiota diversity with OS among patients with allo-HSCT was available from 6 studies [6–9, 44, 45]. Among these studies, 3, 2, and 1 studies detected intestinal microbiota diversity during pre-transplant, post-transplant, and both of period respectively. The overall pooled HR suggested that higher microbiota diversity was associated with OS benefit (HR = 0.66, 95% CI: 0.55–0.78, $p < 0.00001$; $I^2 = 4\%$) (Fig. 2A). Meanwhile, both of the pre-transplant (HR = 0.67, 95% CI: 0.51–0.88, $p = 0.004$; $I^2 = 35\%$) and post-transplant (HR = 0.65, 95% CI: 0.52–0.81, $p = 0.0001$; $I^2 = 0\%$) higher microbiota diversity showed a significant superior OS compared with the lower-diversity group.

Transplant-related mortality

Data to calculate the association of intestinal microbiota diversity with TRM among patients with allo-HSCT was available from 3 studies [6, 7, 45]. Among these studies, 1,

2, and 1 studies detected pre-transplant, post-transplant, and both period of intestinal microbiota diversity respectively. The results revealed that higher microbiota diversity was related to decreased TRM (HR = 0.56, 95% CI: 0.41–0.76, $p = 0.0002$; $I^2 = 0\%$) (Fig. 2B). Meanwhile, both of the pre-transplant (HR = 0.44, 95% CI: 0.22–0.88 $p = 0.02$) and post-transplant (HR = 0.59, 95% CI: 0.42–0.83, $p = 0.003$; $I^2 = 0\%$) higher microbiota diversity showed a significant decreased TRM compared with the lower-diversity group.

Grade II–IV aGVHD

Data to calculate the association of intestinal microbiota diversity with the cumulative incidence of grade II–IV aGVHD among patients with allo-HSCT was available from five studies [7, 41–44], with a total of 476 allo-HSCT patients, of which 190 patients developed grade II–IV aGVHD. The pooled HR suggested that patients with higher microbiota diversity have lower risk of grade II–IV aGVHD (HR = 0.41, 95% CI: 0.27–0.63, $p < 0.0001$; $I^2 = 0\%$) (Fig. 3A). The similar association was found in post-transplant intestinal microbiota diversity (HR = 0.41, 95% CI: 0.26–0.65, $p = 0.0001$; $I^2 = 0\%$). However, Doki et al. [44] suggested that pre-transplant intestinal microbiota diversity failed to reveal a significant difference in the cumulative incidence of grade II to IV aGVHD (HR = 0.44, 95% CI: 0.17–1.14, $p = 0.09$).

cGVHD

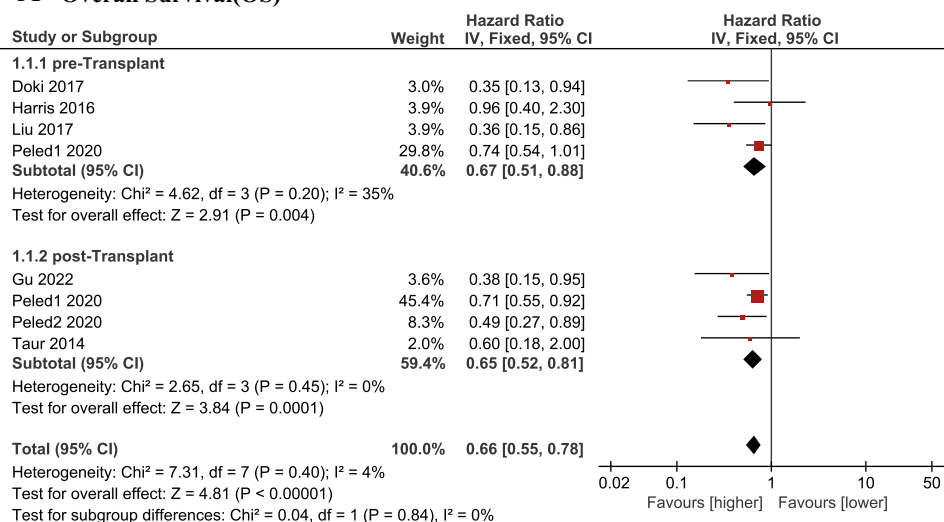
So far, only two studies [7, 46] reported the association between intestinal microbiota diversity and cGVHD incidence, with a total of 147 allo-HSCT patients, of which 39 patients developed cGVHD. Higher microbiota diversity post-transplant was not associated with increased risk of cGVHD (RR = 0.41, 95% CI: 0.09–1.79, $p = 0.24$; $I^2 = 79\%$) (Fig. 3B). The significant heterogeneity may be caused by clinical aspects of the studies, as one study included patients only received anti-thymocyte globulin (ATG)-based myeloablative allo-HSCT, and the other study included patients who received treatment of ATG or not.

Relapse

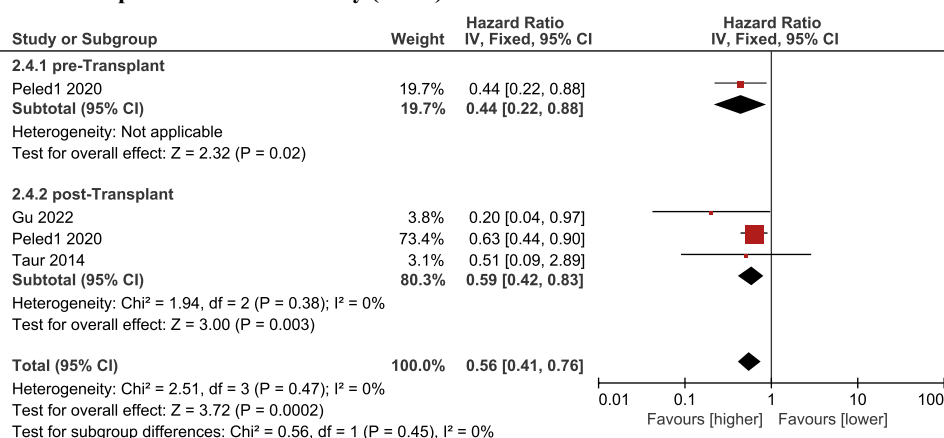
The relationship between intestinal microbiota diversity and relapse was evaluated in three studies [6, 7, 44], with a total of 896 allo-HSCT patients, of which 206 patients relapsed, while no significant association was found between intestinal microbiota diversity and relapse (RR = 1.01, 95% CI: 0.74–1.39, $p = 0.94$; $I^2 = 0\%$) (Fig. 3C).

Fig. 2 Forest plots for intestinal microbiota diversity (higher-diversity group vs. lower-diversity group) and risk of overall survival (OS) (A), transplant-related mortality (TRM) (B)

A Overall Survival(OS)



B Transplant-Related Mortality (TRM)



The association of some specific microbiota abundance with overall survival

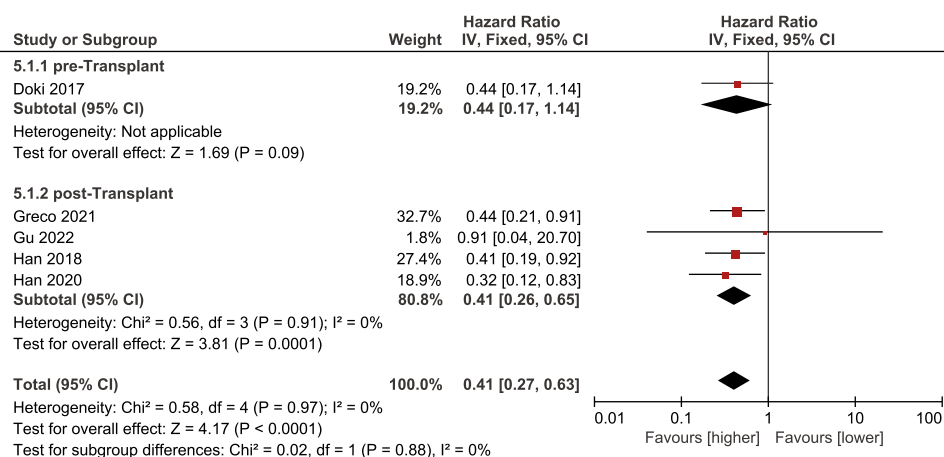
Four microbiotas were eligible for meta-analysis, including *Clostridiales* [14, 15, 49], *Enterococcus* [8, 12, 47, 50], *γ-proteobacteria* [8, 14], and *Candida* [48, 49]. The overall combined estimate suggested that higher abundance of *Clostridiales* was associated with a superior OS (HR = 0.40, 95% CI: 0.18–0.87, $p = 0.02$; $I^2 = 56\%$) (Fig. 4A), while higher abundance of *Enterococcus* (HR = 2.03, 95% CI: 1.55–2.65, $p < 0.00001$; $I^2 = 45\%$) (Fig. 4B), *γ-proteobacteria* (HR = 2.82, 95% CI: 1.53–5.20, $p = 0.0009$; $I^2 = 0\%$) (Fig. 4C), and *Candida* (HR = 3.80, 95% CI: 1.32–10.94 $p = 0.01$; $I^2 = 55\%$) (Fig. 4D) was an adverse prognostic factor for OS.

Sensitivity analysis

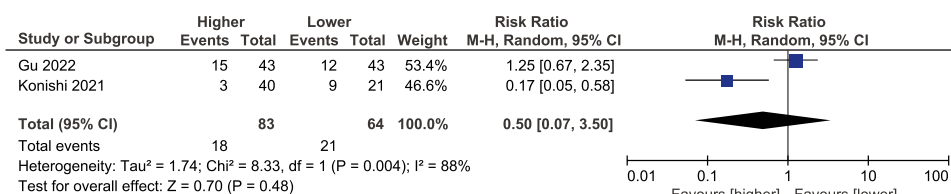
To explore potential source of heterogeneity across studies, we carried out “leave-one-out” sensitivity analysis. The result suggested that overall combined risk estimates were robust except for the association of *Clostridiales* abundance and OS. The pooled estimate for higher abundance of *Clostridiales* with OS was as follows: (HR = 0.40, 95% CI: 0.18–0.87, $p = 0.02$; $I^2 = 56\%$). After exclusion of the study by Malard et al. [49] or Mancini et al. [14] respectively, the overall risk estimate became as follows: (HR = 0.40, 95% CI: 0.12–1.35, $p = 0.14$; $I^2 = 78\%$) and (HR = 0.60, 95% CI: 0.30–1.22, $p = 0.16$; $I^2 = 0\%$). The significant heterogeneity may be caused by different bacteria species and different categories criterion for high

Fig. 3 Forest plots for intestinal microbiota diversity (higher-diversity group vs. lower-diversity group) and risk of grade II–IV aGVHD (A), cGVHD (B), relapse (C)

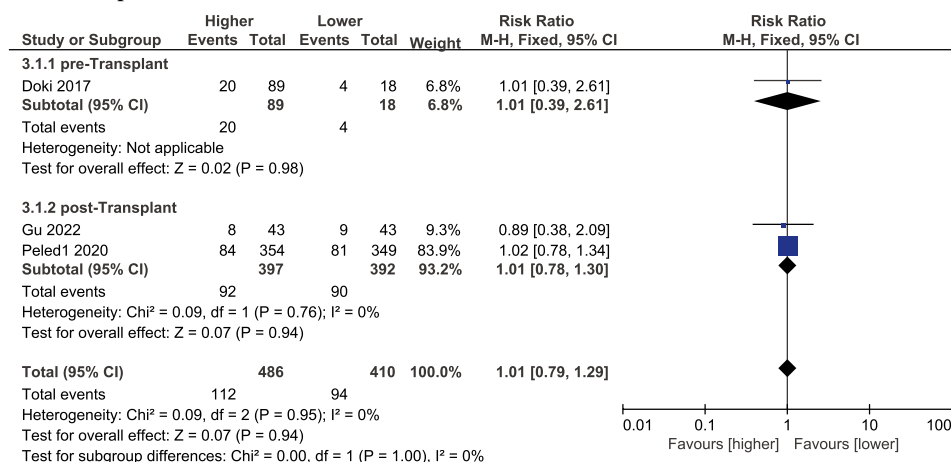
A Grade II–IV aGVHD



B cGVHD



C Relapse



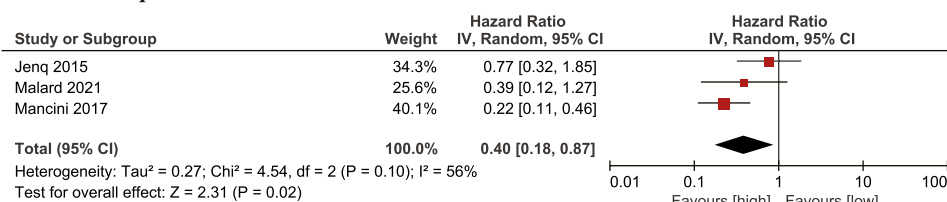
bacteria abundance in these three studies. Jenq et al. [15] reported the association of *Blautia* > 0.05% with OS; Malard et al. [49] reported the association of high *Lachnospiraceae* abundance (higher than median proportion) with OS; Mancini et al. [14] reported the association of *Lachnospiraceae* > 10% with OS. Thus, further studies were needed to improve the quality of evidence and certainty for the association of *Clostridiales* abundance with OS after allo-HSCT.

Discussion

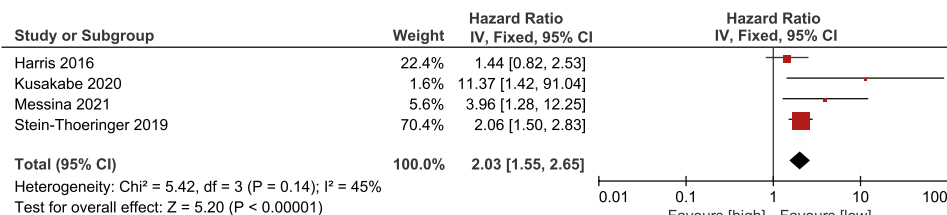
This meta-analysis supports the hypothesis that higher intestinal microbiota diversity was associated with a reduced risk of TRM and aGVHD which contributed to a significantly superior OS. Besides, intestinal microbiota diversity in the period of both pre-transplant and post-transplant significantly associated with OS and TRM. However, no meaningful decrease in the risk of relapse and cGVHD was observed

Fig. 4 Forest plot of prospective cohort studies evaluating the hazard ratios of overall survival (OS) for high abundance of *Clostridiales* (A), *Enterococcus* (B), *γ-proteobacteria* (C), and *Candida* (D)

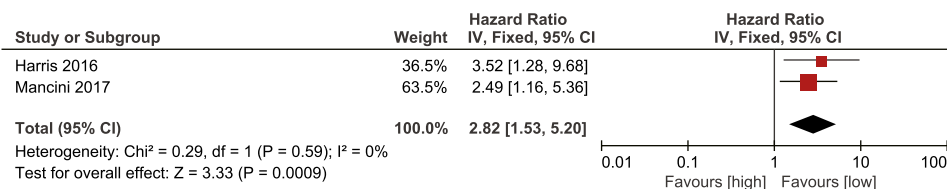
A *Lachnospiraceae*



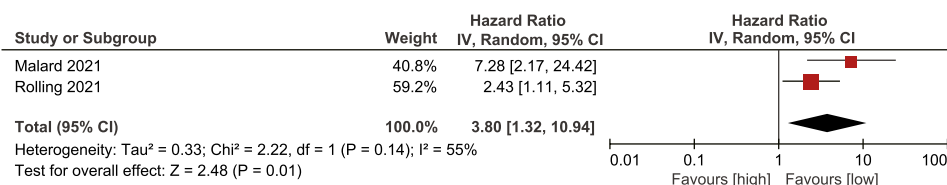
B *Enterococcus*



C *γ-proteobacteria*



D *Candida*



to be associated with intestinal microbiota diversity. More importantly, our meta-analysis suggested that high abundance of *Clostridiales* could be protective factor for OS, while high abundance of *Enterococcus*, *γ-proteobacteria*, and *Candida* was a risk factor for OS after allo-HSCT.

Results in relation to other studies

Nowadays, only one relevant meta-analysis reported that higher intestinal microbiota diversity presented increased OS (RR = 1.58 (1.19–2.09), $p = 0.002$, $I^2 = 79\%$) and lower risk of TRM (RR = 0.45 (0.26–0.76), $p = 0.003$, $I^2 = 53\%$) [51]. However, their result showed statistically significant heterogeneity. In our meta-analysis, we added more relevant prospective studies, and the pooled HR suggested that higher microbiota diversity was associated with OS benefit (HR = 0.66, 95% CI: 0.55–0.78, $p < 0.00001$; $I^2 = 4\%$), as

well as decreased TRM (HR = 0.56, 95% CI: 0.41–0.76, $p < 0.0002$; $I^2 = 0\%$), without significant heterogeneity. Besides, the sensitivity analysis suggested that the combined risk estimates were robust. What's more, our result highlighted the effects of microbiota diversity at two specific timepoint, before HSCT (pre-transplant) and after HSCT (post-transplant). By subgroup analysis, higher intestinal microbiota diversity during both pre-transplant and post-transplant periods showed significant protective role on OS and TRM after allo-HSCT. Previous studies have proved that patients arrived for transplantation with microbiota compositions that were already distinct from those of healthy volunteer and that were characterized by loss of diversity and domination by single taxa [6, 9, 10], and the decrease in microbial diversity observed after transplantation persists for up to 1 year after transplantation [52]. Montassier et al. showed that the risk of bloodstream infections could be predicted by microbiota composition before

HSCT [53] and that the risk of TRM could be predicted by the timing of exposure to antibiotics and by colonization with antibiotic-resistant bacteria before transplantation [54–57]. Thus, microbiota diversity can be early predictor of OS and TRM for allo-HSCT patients. Interventions, aiming to prevent microbiota injury and restore microbiota diversity, need to be taken seriously both before and during allo-HSCT.

It should be noted that our study was the first meta-analysis to estimate the association of intestinal microbiota diversity with aGVHD in a systematic way, providing a more comprehensive knowledge of this field. The pooled HR suggested that patients with higher microbiota diversity during post-transplant have lower risk of grade II–IV aGVHD (HR = 0.41, 95% CI: 0.26–0.65, $p = 0.0001$; $I^2 = 0\%$). However, previous studies found that pre-transplant intestinal microbiota diversity failed to reveal a significant difference in the incidence aGVHD [14, 44]. Two other meta-analysis [51, 58] showed that patients receiving gut decontamination (GD) were at higher risk of aGVHD, as well as gastrointestinal aGVHD, suggesting that GD during allo-HSCT may destroy the diversity of the intestinal microbiota and increase the risk of aGVHD, consistent with our findings. Our meta-analysis suggested that microbiota diversity during post-transplant could be more suitable as a predictor of aGVHD than pre-transplant microbiota diversity.

Moreover, this meta-analysis showed that higher microbiota diversity was not statistically associated with increased risk of cGVHD (RR = 0.41, 95% CI: 0.09–1.79, $p = 0.24$; $I^2 = 79\%$), although the heterogeneity was significant. Previous studies indicate the decrease in overall gut diversity as a result of antibiotic therapy subsequently increases the risk of gastrointestinal aGVHD [23, 59], but not cGVHD diagnosis or cGVHD mortality [60], which is consistent with our result that higher intestinal microbiota diversity was significantly related to a lower risk of grade II–IV aGVHD, nor cGVHD. Besides, a case–control study [52] showed that intestinal microbiota diversity between cGVHD patients and controls has no significant differences at the different time points (pre-transplant, periengraftment, and peri-day 100 windows), correlated to our results, while they found that short-chain fatty acid (SCFA) metabolism in stool samples between cGVHD patients and controls was significantly different [52]. So, we suggested that, compared with intestinal microbiota diversity, microbiota metabolism could be more novel and practical target for prevention and treatment of cGVHD.

Last but not least, this meta-analysis enrolled relevant studies about the association between some specific microbiota and outcomes of allo-HSCT. The overall combined estimate suggested that higher abundance of *Clostridiales* was associated with a superior OS (HR = 0.40, 95% CI: 0.18–0.87, $p = 0.02$; $I^2 = 56\%$), while higher abundance of

Enterococcus (HR = 2.03, 95% CI: 1.55–2.65, $p < 0.00001$; $I^2 = 45\%$), *γ-proteobacteria* (HR = 2.82, 95% CI: 1.53–5.20, $p = 0.0009$; $I^2 = 0\%$), and *Candida* (HR = 3.80, 95% CI: 1.32–10.94 $p = 0.01$; $I^2 = 55\%$) were adverse prognostic factors for OS. However, due to limited availability of eligible data, we refrained from performing subgroup analysis differentiating between the pre-transplant period and the post-transplant period, concerning the correlation between the abundance of specific intestinal microbiota subsets and overall survival after allo-HSCT. In the study by Messina et al. [47], they found that patients with *Enterococcus* domination had reduced overall survival, and they defined “*Enterococcus* domination” as “at least one stool sample with *Enterococcus* $\geq 30\%$ during day -30 up to day + 180”, indicating that “*Enterococcus* domination” happened during both post-transplant period and pre-transplant period may associated with overall survival after allo-HSCT, while further detailed studies were needed to draw a definite conclusion in this field. Researchers suggested that broad-spectrum antibiotic treatments are the likely risk to develop gastrointestinal aGVHD, while limited-spectrum activity antibiotics are not associated with gastrointestinal aGVHD-related mortality [61]. A reasonable explanation for this is that some specific bacterial species have characteristics beneficial to the gut, while others cause harm if given the opportunity to growth [62]. Non-antibiotic-based therapeutic strategy such as a lactose-free diet attenuated the outgrowth of pathobionts like *Enterococcus* and possibly improved clinical outcomes by modulating dietary sources of nutrients for pathogenic bacteria [12]. Our meta-analysis further provided evidence of microbiome markers for OS, which offering new tools for individualized preemptive or therapeutic strategies to improve allo-HSCT outcomes.

Limitations of the review

In addition to strengths described above, our meta-analysis also has certain limitations. First, potential confounders included for the risk estimates adjustment differed among the eligible studies, even some data were unadjusted. Second, there were differences in measurement of microbiota diversity index across studies, and the categories’ criterion for high-diversity and lower-diversity groups was according to the median diversity value, which varies across studies. Third, insufficient quantity of enrolled studies are possible causes of the observed heterogeneity, although we addressed this heterogeneity by applying the random-effects model in the analysis. At last, possible publication bias that we cannot address statistically might challenge the robustness of our conclusions. Besides, only studies published in English were included, and some high-quality articles published in other

languages might have been excluded, which would increase the publication bias.

Conclusions

This meta-analysis provides further evidence that higher intestinal microbiota diversity was associated with a lower risk of TRM, aGVHD, and superior OS. However, no meaningful decrease in the risk of relapse and cGVHD was observed to be associated with intestinal microbiota diversity. Higher abundance of *Clostridiales* could be protective factor for OS, while high abundance of *Enterococcus*, *γ-proteobacteria*, and *Candida* was a risk factor for OS after allo-HSCT. Our meta-analysis will be a useful tool for clinical practice and guidance of individualized preemptive or therapeutic strategies to improve allo-HSCT outcomes. More studies with detailed assessment of the effect of specific bacterial or fungus using standardized methods were needed to draw a definite conclusion in the field.

Abbreviations *allo-HSCT*: Allogeneic hematopoietic stem cell transplantation; *GVHD*: Graft-versus-host disease; *aGVHD*: Acute graft-versus-host disease; *cGVHD*: Chronic graft-versus-host disease; *HR*: Hazard ratio; *RR*: Risk ratio; *CI*: Confidence intervals; *TRM*: Transplant-related mortality; *OS*: Overall survival; *MeSH*: Medical Subject Headings; *RevMan*: Review Manager; *ATG*: Anti-thymocyte globulin; *GD*: Gut decontamination; *CR*: Complete remission; *FMT*: Fecal microbiota transplantation; *IMD*: Intestinal microbiota diversity

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00277-023-05460-7>.

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Author contribution WSF, YXY, and XHW contributed to conception and design. WSF and CX conducted the study search and selection. WSF, CHQ, CX, and ZHY conducted the statistical analysis. WSF, YXY, and ZHY drafted the manuscript, which was critically revised for important intellectual content by XHW. HLN, PWJ, and ZXJ prepared figures. All authors reviewed and approved the final version. XHW is guarantor.

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Data availability Data available within the article or its supplementary materials.

Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Competing interests The authors declare no competing interests.

Disclaimer The funders had no role in study design, data collection and analysis, preparation of the manuscript, or decision to publish.

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